Customer Segmentation and Churn Pattern Analytics in European Banking: A Data-Driven Dashboard Approach

Pallavi Modi
jpallavi6@gmail.com

***Abstract*—**Customer attrition, commonly referred to as churn, is one of the most pressing challenges facing retail banks, since retaining an existing customer is consistently less costly than acquiring a new one. This paper presents an end-to-end analytics pipeline for exploring churn behavior in a European banking dataset comprising 10,000 customer records spanning France, Germany, and Spain. The pipeline covers data cleaning, feature-based customer segmentation (by age, tenure, credit score, and account balance), exploratory statistical analysis, and the deployment of an interactive Streamlit dashboard for real-time churn monitoring. The analysis reveals an overall churn rate of 20.37%, with pronounced disparities across geography, age, product holding, and account-activity status: German customers, older customers, disengaged (inactive) members, and customers holding three or more products exhibit substantially elevated churn risk. These findings are synthesized into actionable segments that can support targeted retention strategies. The resulting dashboard allows stakeholders to filter, drill down, and export customer segments, translating exploratory analytics into an operational decision-support tool.

Index Terms*—customer churn, customer segmentation, banking analytics, exploratory data analysis, data visualization, Streamlit dashboard.*

I. INTRODUCTION

Customer churn, the phenomenon in which a client discontinues their relationship with a service provider, is a first-order concern for retail banks operating in mature, competitive markets. Acquiring a new customer typically costs several times more than retaining an existing one, which means even modest reductions in churn rate can yield disproportionate gains in lifetime value. Despite banks accumulating rich, customer-level transactional and demographic data, this information is often under-utilized: churn-management strategies remain generic and reactive rather than data-driven and targeted.

This work addresses three concrete objectives. First, to quantify the overall churn rate and decompose it across customer segments defined by geography, demographics, and account characteristics. Second, to identify which segments carry disproportionate churn risk, with particular attention to high-value (high-balance) customers whose departure carries outsized financial impact. Third, to translate these exploratory findings into an interactive, filterable dashboard that a non-technical stakeholder — such as a retention or relationship-management team — can use without writing code.

The remainder of this paper is organized as follows. Section II describes the dataset. Section III details the data-preparation and segmentation methodology. Section IV presents the empirical results. Section V describes the interactive dashboard implementation. Section VI discusses the findings and their business implications, and Section VII concludes with limitations and directions for future work.

II. DATASET DESCRIPTION

The analysis uses a bank-customer record set of 10,000 rows and 14 analytically relevant attributes after the removal of identifier columns. Each row represents one customer and includes demographic attributes (Age, Gender, Geography), account attributes (CreditScore, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary), and the binary target label Exited, which indicates whether the customer churned (1) or was retained (0). The dataset contains no missing values and no duplicate records, so no imputation or deduplication was required.

Customers are drawn from three European markets — France (5,014 customers), Germany (2,509 customers), and Spain (2,477 customers) — with a near-even gender split (5,457 male, 4,543 female). Table I summarizes the descriptive statistics of the principal numeric attributes.

Attribute	Mean	Std. Dev.	Min	Max
Age (years)	38.92	10.49	18	92
Credit Score	650.53	96.65	350	850
Tenure (years)	5.01	2.89	0	10
Balance (€)	76,486	62,397	0	250,898
Est. Salary (€)	100,090	57,510	12	199,992

TABLE I. DESCRIPTIVE STATISTICS OF NUMERIC ATTRIBUTES (N = 10,000)

III. METHODOLOGY

A. Data Cleaning

The raw dataset was first stripped of non-predictive identifier columns (CustomerId, Surname) that carry no analytical signal and risk leaking spurious identity information into downstream analysis. The dataset was then checked column-by-column for null values and duplicate rows; none were found, confirming the data was analysis-ready without further remediation.

B. Feature-Based Segmentation

To support interpretable, business-relevant comparisons, four continuous attributes were discretized into categorical segments:

- Age Group: <30, 30–45, 46–60, and 60+ years.
- Credit Band: Low (<500), Medium (500–699), and High (≥700).
- Tenure Group: New (≤3 years), Mid-term (4–7 years), and Long-term (>7 years).
- Balance Segment: Zero Balance (=0), Low Balance (0–100,000), and High Balance (>100,000).

These derived segments allow churn rate to be examined as a categorical cross-tabulation rather than a continuous correlation, which is more directly actionable for a retention team designing segment-specific interventions.

C. Exploratory Analysis

Churn rate was computed as the mean of the binary Exited indicator within each segment, expressed as a percentage. Group-wise churn rates were computed for geography, gender, age group, credit band, tenure group, balance segment, product count, credit-card ownership, and account-activity status. A dedicated "high-value customer" cohort was defined as customers with an account balance exceeding €100,000, reflecting the segment whose attrition carries the greatest financial consequence. All group comparisons were implemented in Python using pandas for aggregation and Matplotlib/Seaborn for static visualization during the exploratory phase, prior to reproducing the key views as interactive Plotly charts in the deployed dashboard.

IV. RESULTS AND ANALYSIS

A. Overall Churn Rate

Of the 10,000 customers in the dataset, 2,037 churned, yielding an overall churn rate of 20.37%. This baseline rate serves as the reference point against which segment-level rates are compared in the remainder of this section.

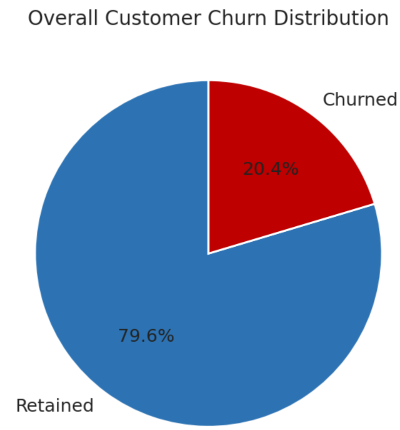

Fig. 1. Overall distribution of retained vs. churned customers.

B. Geographic Analysis

Churn is highly uneven across markets. Germany exhibits a churn rate of 32.44%, roughly double that of France (16.15%) and Spain (16.67%), despite Germany accounting for only about a quarter of the customer base. This concentration suggests market-specific factors — such as local competition, product fit, or service quality — are driving disproportionate attrition in Germany, making it a priority market for retention intervention.

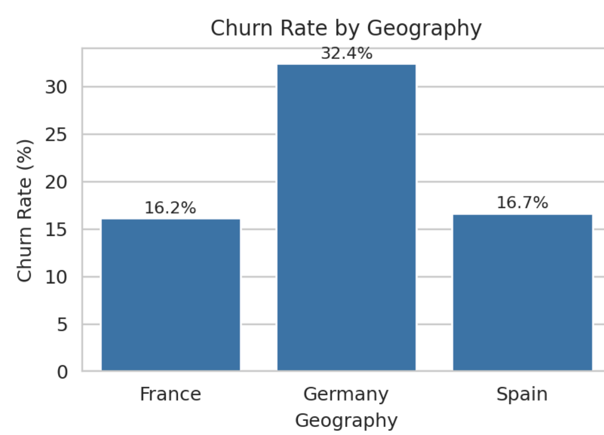

Fig. 2. Churn rate by geography.

C. Demographic Analysis

Age is strongly associated with churn, but not monotonically: churn rises from 7.52% for customers under 30 to a peak of 51.12% for the 46–60 age band, before falling back to 24.78% for customers over 60. The 46–60 segment therefore represents the single highest-risk age cohort and warrants targeted retention outreach. Gender differences are more modest but still material: female customers churn at 25.07% versus 16.46% for male customers, an 8.6 percentage-point gap.

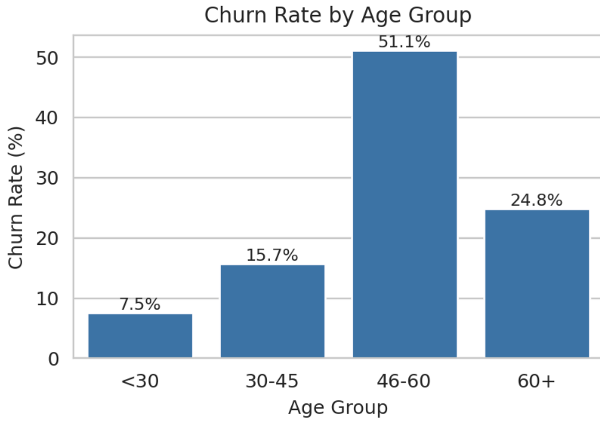

Fig. 3. Churn rate by age group.

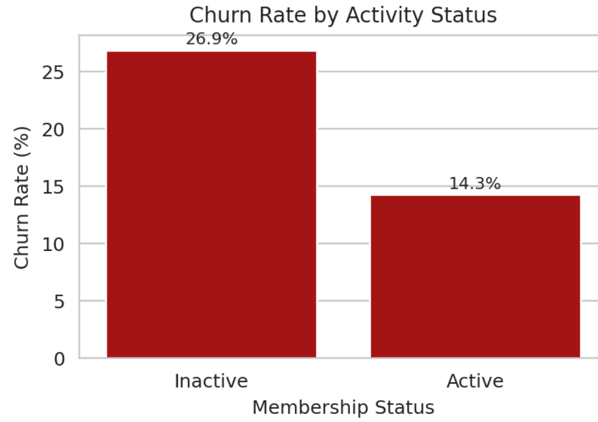

Fig. 5. Churn rate by account-activity status.

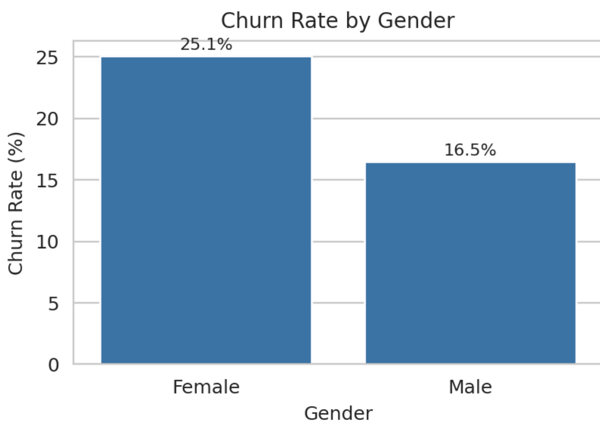

Fig. 4. Churn rate by gender.

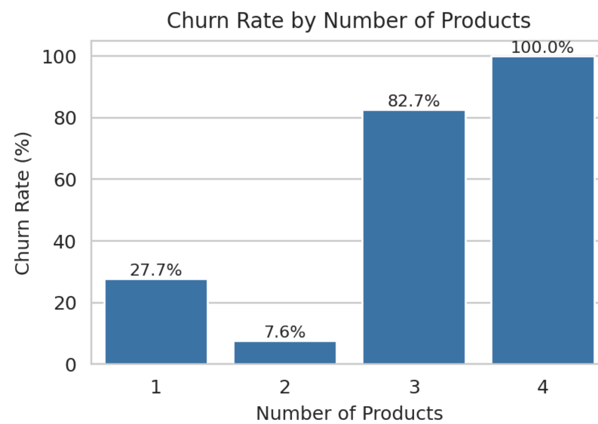

Fig. 6. Churn rate by number of products held.

D. Engagement and Product Holding

Account-activity status is one of the strongest single predictors of churn observed in this analysis: inactive members churn at 26.85% compared to 14.27% for active members, a gap of roughly 12.6 percentage points. Product holding shows an even sharper, non-linear relationship: churn is 27.71% for customers with a single product, falls to a low of 7.58% for customers with two products, and then rises sharply to 82.71% for customers with three products and 100% for the small cohort ($n = 60$) holding four products. This U-shaped pattern indicates that moderate product cross-sell (two products) is protective, while over-selling beyond two products is a strong distress signal rather than a sign of deeper relationship engagement. Credit-card ownership, by contrast, shows negligible association with churn (20.81% without a card vs. 20.18% with one).

E. High-Value Customer and Balance Segment Analysis

Customers are also differentiated by account balance. Zero-balance customers churn least (13.82%), while high-balance customers (>€100,000) churn most (25.23%), with low-balance customers falling in between (20.58%). The high-value cohort (balance > €100,000) comprises 4,799 customers — nearly half the customer base — and its 25.23% churn rate, applied against substantial account balances, represents the largest concentration of financial risk in the portfolio. This finding motivated the dedicated high-value customer explorer built into the dashboard, described in Section V.

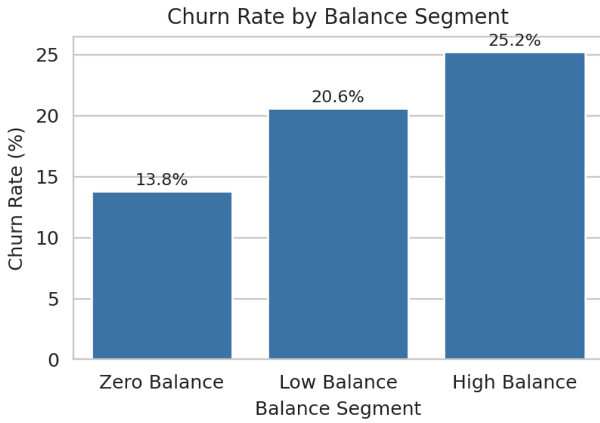

Fig. 7. Churn rate by account-balance segment.

F. Summary of Segment-Level Churn Rates

Segment	Category	Churn Rate	n
Geography	Germany	32.44%	2,509
Geography	Spain	16.67%	2,477
Geography	France	16.15%	5,014
Age Group	46–60	51.12%	—
Age Group	<30	7.52%	—
Products	3 products	82.71%	266
Products	2 products	7.58%	4,590
Activity	Inactive	26.85%	—
Activity	Active	14.27%	—
Balance	High (>€100k)	25.23%	4,799
Balance	Zero	13.82%	—
Overall	All customers	20.37%	10,000

TABLE II. CHURN RATE BY KEY SEGMENT (SORTED WITHIN EACH CATEGORY)

V. INTERACTIVE DASHBOARD IMPLEMENTATION

The exploratory findings were operationalized as an interactive web dashboard built with Streamlit and Plotly, enabling non-technical stakeholders to explore churn patterns without writing code. On load, the application reads the raw customer table, drops identifier columns, and recomputes the age, balance, and tenure segments described in Section III, along with a human-readable churn-status label.

The interface exposes sidebar filters for geography, gender, age group, and tenure group, which dynamically subset all downstream visualizations. A key-performance-indicator row surfaces total customers, churned customers, churn rate, average balance, and average salary for the current filter selection. Below the KPI row, the dashboard presents: (i) an overall churn-distribution donut chart alongside a geography-wise churn bar chart; (ii) side-by-side box plots comparing the age and tenure distributions of churned versus retained customers; (iii) a high-value customer explorer with an adjustable minimum-balance slider and a jittered box plot showing balance distribution by churn status, with geography, gender, and age exposed on hover; (iv) a segment-level churn bar chart across age groups; and (v) a drill-down data table filterable by country, with a button to export the currently filtered records as CSV.

This design mirrors a common analytics-to-decision-support pattern: static exploratory notebooks establish which segments matter, and the deployed dashboard then makes those segments explorable in real time, letting a retention manager, for instance, isolate inactive, high-balance German customers between ages 46 and 60 in a few clicks.

VI. DISCUSSION

Several actionable patterns emerge from the analysis. First, churn risk in this portfolio is concentrated rather than diffuse: three factors — German geography, mid-to-late-career age (46–60), and holding three or more products — are each associated with churn rates two to four times the portfolio baseline, and customers exhibiting more than one of these traits simultaneously likely represent the highest-priority retention targets. Second, account-activity status is a strong and readily actionable signal: because “inactive” is a behavioral flag a bank can monitor continuously, it is well suited to triggering automated early-warning retention workflows rather than relying solely on periodic segment reporting. Third, the non-monotonic relationship between product count and churn cautions against a simplistic “more cross-sell is always better” strategy; beyond two products, additional product holding appears to coincide with customer distress (e.g., a customer opening additional accounts shortly before leaving) rather than the deeper engagement typically assumed to reduce churn.

From a financial-risk perspective, the elevated churn rate among high-balance customers (25.23%) is particularly consequential, since this cohort disproportionately determines the bank's deposit base and fee income. Prioritizing retention spend on this segment, rather than distributing it uniformly across the customer base, is likely to yield a substantially higher return on retention investment.

VII. CONCLUSION AND FUTURE WORK

This paper presented a complete churn-analytics pipeline for a 10,000-customer European banking dataset, combining data cleaning, rule-based segmentation, exploratory statistical analysis, and an interactive Streamlit/Plotly dashboard. The analysis found an overall churn rate of 20.37%, with Germany, the 46–60 age band, inactive members, and customers holding three or more products emerging as the highest-risk segments, and high-balance customers representing the segment of greatest financial consequence.

The present work is descriptive rather than predictive: it characterizes association, not causation, and does not yet estimate individual-level churn probability. Natural extensions include training a supervised classification model (e.g., logistic regression, gradient-boosted trees) to produce a per-customer churn-risk score for proactive outreach; incorporating this score into the dashboard as a ranked "at-risk customer" worklist; and conducting a formal uplift analysis to quantify which retention interventions are most effective for each identified segment.

REFERENCES

[1] W. McKinney, "Data Structures for Statistical Computing in Python," in Proc. 9th Python in Science Conf. (SciPy), 2010, pp. 56–61.

[2] J. D. Hunter, "Matplotlib: A 2D Graphics Environment," Computing in Science & Engineering, vol. 9, no. 3, pp. 90–95, 2007.

[3] M. L. Waskom, "seaborn: statistical data visualization," Journal of Open Source Software, vol. 6, no. 60, p. 3021, 2021.

[4] Plotly Technologies Inc., "Plotly Python Open Source Graphing Library," plotly.com, 2015.

[5] Streamlit Inc., "Streamlit Documentation," docs.streamlit.io, 2024.

[6] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[7] "Bank Customer Churn Modelling Dataset," Kaggle, [Online dataset]. Note: also distributed under the filename Churn_Modelling.csv in this study.